The Journal of Open Source Software

cifkit: A user-friendly Python package for high-throughput CIF analysis

Sangjoon Lee [1¶] **and Anton O. Oliynyk** [2,3]

1 Department of Applied Physics and Applied Mathematics, Columbia University, New York, NY 10027, USA **2** Department of Chemistry, Hunter College, City University of New York, New York, NY 10065, USA **3** Ph.D. Program in Chemistry, The Graduate Center of the City University of New York, New York, NY 10016, USA ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

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Editor: Rachel Kurchin

Reviewers:

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Submitted: 30 August 2024
Published: unpublished

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Summary

`cifkit` is designed to provide a set of intuitive utility functions and variables for processing large datasets, on the order of tens of thousands of .cif files. `cifkit` serves as an engine for building Python applications that automate crystal structure analysis, enabling the extraction of physics-based information crucial for understanding geometric configurations and identifying irregularities. `cifkit` also offers various tools for determining coordination numbers, plotting coordination geometry-based polyhedron from each site, calculating bond fractions, moving and copying .cif files based on a set of attributes, and determining atomic mixing information.

Statement of need

In solid state chemistry and materials science, the Crystallographic Information File (CIF) (Hall et al., 1991) is the predominant file format used to store and distribute crystal structure information. There are open-source Python packages that read, edit, and create CIF files. Python Materials Genomics (pymatgen) (Ong et al., 2013) offers functionalities beyond the aforementioned features, such as generating electronic structure properties and phase diagrams. Similarly, the Atomic Simulation Environment (ASE) (Larsen et al., 2017) provides a suite of powerful tools for generating and running atomistic simulations. However, both projects are tailored for users with prior programming experience, often necessitating the use of the tools with API documentation.

Given that .cif files are primarily generated by experimentalists, often with the help of software, `cifkit` is designed for users with limited programming backgrounds to lower the entry barrier for the experimentalist community. This tool allows members to analyze their synthesized materials using numerical descriptions, including complex tasks such as extracting descriptors for geometry-based polyhedra from each atomic site. `cifkit` helps systematically describe materials and automatically extract intuitive, physics-based, and measurable information from .cif files, tasks that may require time-consuming manual work with GUI-based tools like VESTA, Diamond, and CrystalMaker. Furthermore, `cifkit` offers unique functionalities to visualize the distribution of file counts based on attributes, extract atomic mixing information at the bond pair level, preprocess .cif files to parse the atomic site element from the atomic site label, and identify and separate ill-formatted files in a few lines of code. By simplifying user interactions while maintaining robust functionality, `cifkit` enables a broader range of scientists to leverage computational tools in their research.

Examples

The following example code demonstrates the user-friendly nature of the tool for users with limited programming experience. The full installation process can be executed via a Jupyter notebook, which is distributed through the Google Colab URL provided in the official documentation.

```
from cifkit import Cif, Example

# Initalize with the .cif file path
cif = Cif(Example.Er10Co9In20_file_path)
cif.formula
cif.structure
cif.unique_elements
cif.unitcell_lengths
cif.unitcell_angles
```

To extract information from a set of .cif files:

```
from cifkit import CifEnsemble, Example

# Initalize with the folder path containing .cif files
ensemble = CifEnsemble(Example.ErCoIn_big_folder_path)
ensemble.unique_formulas
ensemble.unique_structures
ensemble.unique_elements

# Determine shortest pair distance per .cif file
ensemble.minimum_distances

# Filter .cif by formula
ensemble.filter_by_formulas(["LaRu2Ge2"])

# Filter .cif by space group name
ensemble.filter_by_space_group_names("Im-3m")
```

Applications

`cifkit` serves as a core package for building applications used by academic and national laboratories for crystal structural analysis and machine learning studies. CIF Bond Analyzer (CBA) utilizes `cifkit` to extract coordination geometry information for newly a discovered phase (Tyvanchuk et al., 2024). The Structure Analysis/Featurizer (SAF) employs `cifkit` to construct and extract physics-based geometric features for binary and ternary compounds. Furthermore, geometric features generated with `cifkit` are being incorporated into a follow-up study on thermoelectric materials (Barua et al., 2024), building upon the compositional properties explored in (Lee et al., 2024).

Testing and documentation

97 percent of the code is covered according to Codecov. The documentation is provided at https://bobleesj.github.io/cifkit.

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Acknowledgement

We acknowledge the initial testing done by Nishant Yadav, Siddha Sankalpa Sethi, and Arnab Dutta from the Indian Institute of Technology, Kharagpur. We also thank Emil Jaffal, Danila Shiryaev, and Alex Vtorov from CUNY Hunter College for testing. We acknowledge Fabian Zills for his recommendations on Python tooling.

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